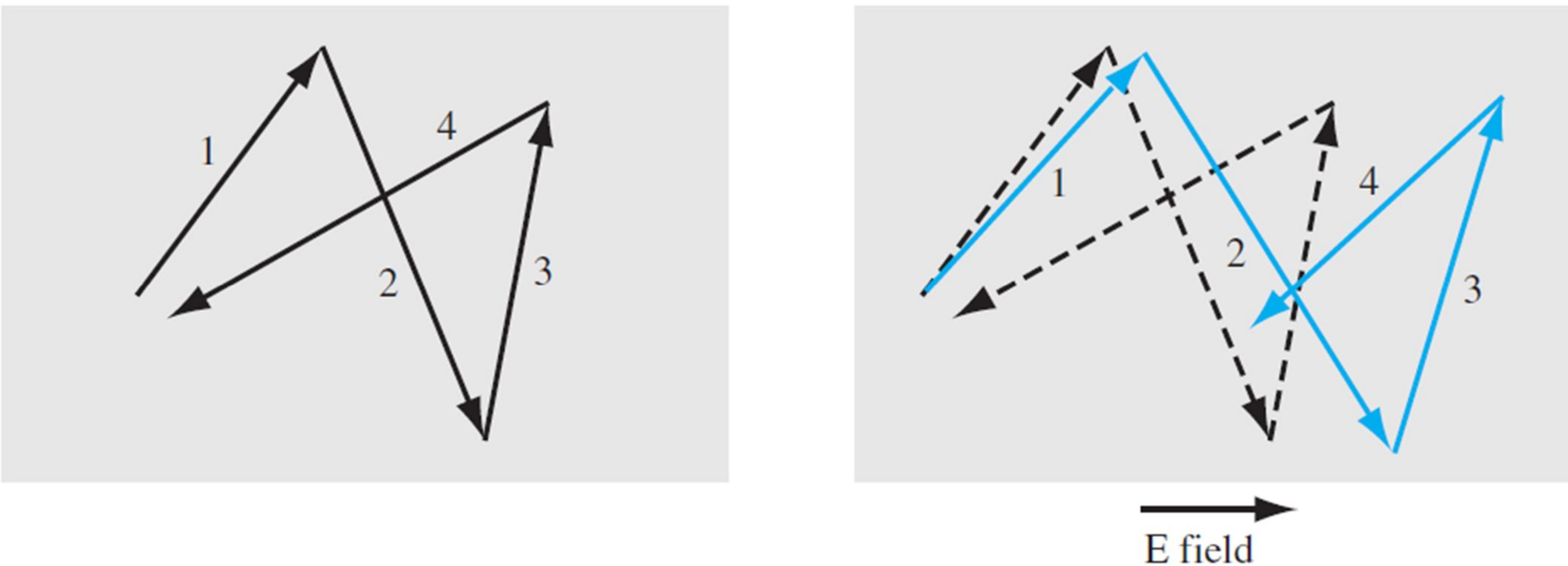
L5: Carrier Transport Phenomena



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Topics & objective

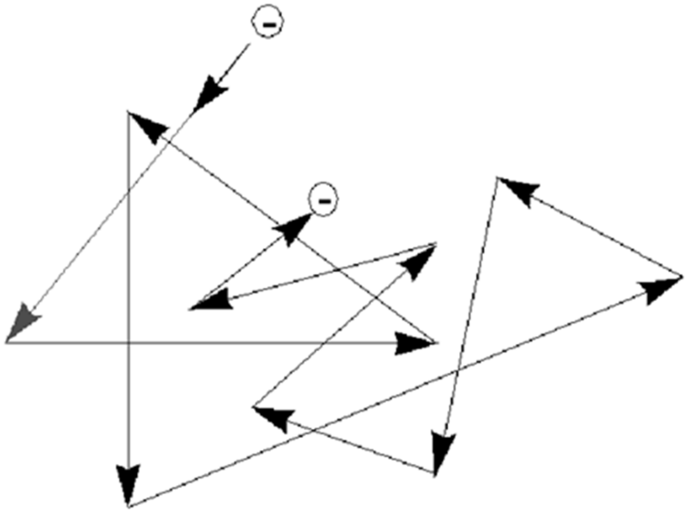
$$\boldsymbol{J = nqv = nquE}$$

- In last chapter, we already study the carrier density n
- In this chapter, we will study the transport mechanism of carrier, i.e., the effect of external fields (electric field, magnetic field) on the movement of carriers.

- Discuss drift and diffusion current densities
- Explain why carriers reach an average drift velocity
- Discuss mechanism of lattice and impurity scattering
- Define mobility, conductivity and resistivity
- Discuss mobility dependence on temperature and impurity and velocity saturation
- State the Einstein relation
- Describe the Hall effect

Thermal Motion

Even in the absence of E-field the electrons have **random** thermal velocity (v_{th}), but net current is zero due to random movement.



$$\textit{Average electron or hole kinetic energy} = \frac{3}{2}kT = \frac{1}{2}mv_{th}^2$$

$$v_{th} = \sqrt{\frac{3kT}{m_{eff}}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23}\,\mathrm{JK^{-1}} \times 300\mathrm{K}}{0.26 \times 9.1 \times 10^{-31}\,\mathrm{kg}}}$$

$$= 2.3 \times 10^5\,\mathrm{m/s} = 2.3 \times 10^7\,\mathrm{cm/s}$$

Mean free path

Average distance between collision = mean free path (l_m)

$l_m \approx$ 10 nm to 1 μm in Si

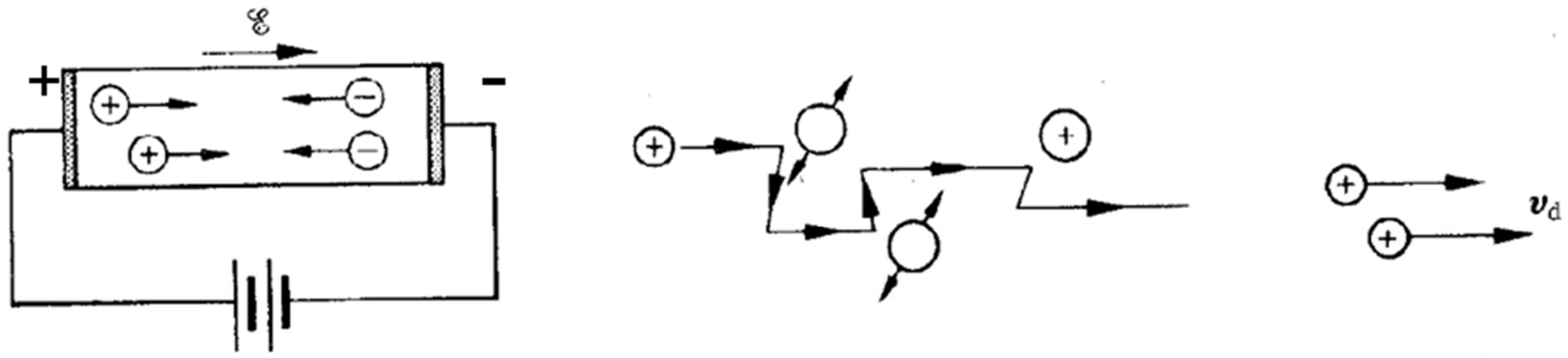
Dependent on material properties such as crystal structure

Since $v_{th} \approx 10^7$ cm/sec, the mean free time between collisions

$$\tau_{cp} \approx \frac{10^{-5} cm}{10^{7} cm/\sec} \approx 1\, p\sec$$

Drift (Electric Field Induced Motion)

When an electric field (E) is applied across the crystal, each electron is acted upon by a force –qE by the field. This force may not be enough to alter the motion of an individual electron. However, when averaged over all electrons there is a net motion of the group.

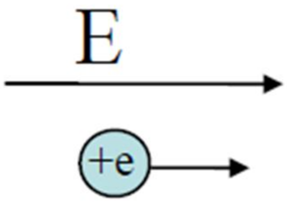


(a) Motion of holes in a Semiconductor bar

(b) Motion on a microscopic scale

(c) Motion on a macroscopic scale

Velocity of the particles

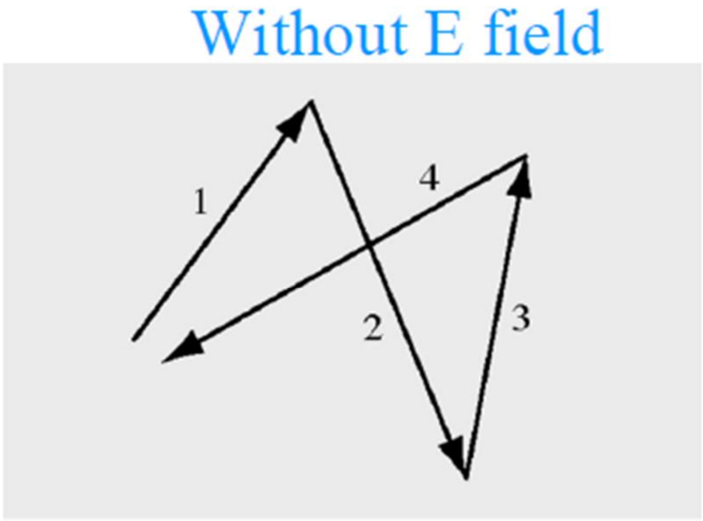


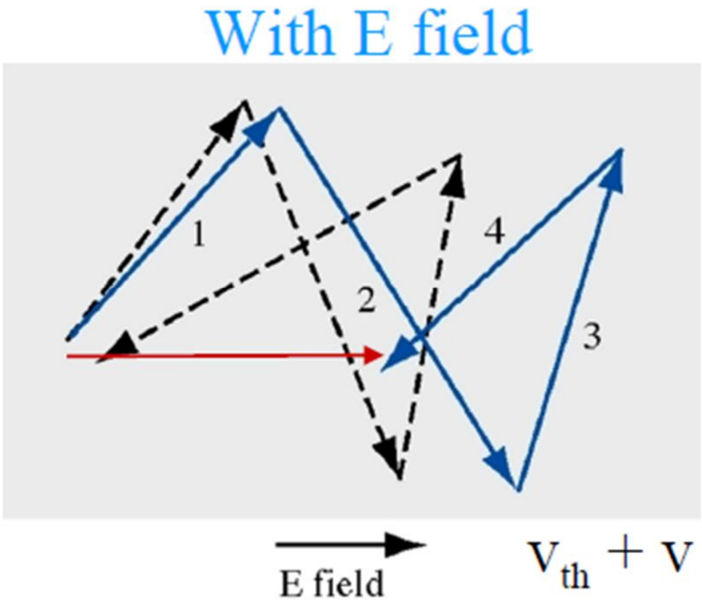
$$F = m_p^* \frac{dv}{dt} = e\mathrm{E} \qquad v = \frac{e\mathrm{E}t}{m_p^*}$$

v → drift velocity of the hole in the electric field

So does velocity monotonically increase with time?

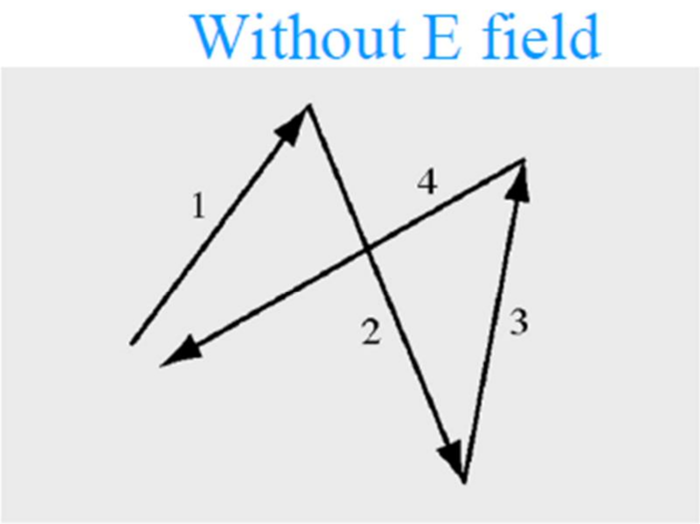
Thermal and drift velocities

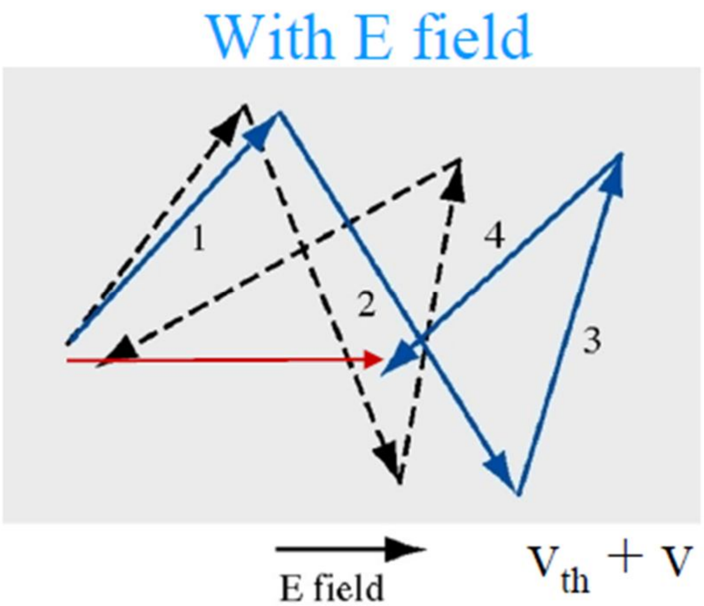




- Even in the absence of E-field the holes have random thermal velocity (v_{th})
- They collide with ionized impurity atoms and thermally vibrating lattice atoms.
- Let τ_{cp} mean time between collisions.
- with E-field: net drift of holes in the direction of the E-field
- net drift velocity is small perturbation on random thermal velocity.
- so τ_{cp} remains almost unchanged even in the presence of E-field.

Thermal and drift velocities

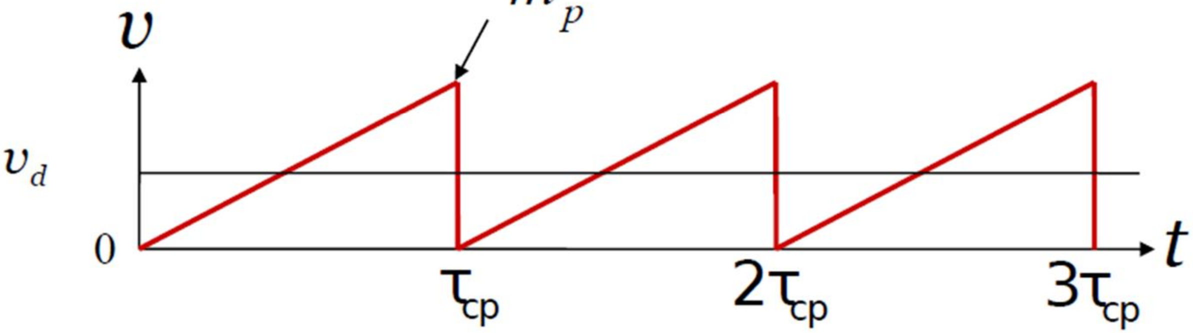


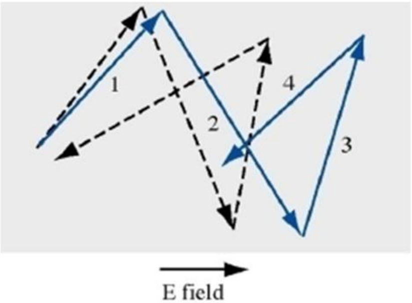


Now, $\upsilon = \dfrac{eEt}{m_p^*}$ $\Rightarrow$ $v_{peak} = \dfrac{e\tau_{cp}}{m_p{}^*}E$

Drift velocity v_d

$$v_{peak} = \frac{e\tau_{cp}}{m_p^*}E$$





(b)

Average drift velocity = $\langle v_d \rangle = \frac{1}{2}\left(\frac{e\tau_{cp}}{m_p^*}\right)E$

Using more accurate model including the effect of statistical distribution,

$$\langle v_d \rangle = \underbrace{\boxed{\left(\frac{e\tau_{cp}}{m_p^*}\right)}}_{\mu_p \text{ (mobility)}} E$$

Mobility

$$\boxed{\upsilon_d = \mu \mathrm{E}}$$

For holes: μ_p → *hole mobility*

$$\boxed{J_{p|drf} = (ep)\upsilon_{dp} = e\mu_p p\mathrm{E}}$$

For electrons: μ_n → *electron mobility* $\upsilon_{dn} = -\mu_n \mathrm{E}$

$$\boxed{J_{n|drf} = (-en)(-\mu_n \mathrm{E}) = e\mu_n n\mathrm{E}}$$

Mobility

$$\mu = \upsilon_d / \mathbf{E} \qquad \text{Unit: cm}^2/\text{Vs}$$

	μ_n (cm²/V-s)	μ_p (cm²/V-s)
Silicon	1350	480
Gallium Arsenide	8500	400
Germanium	3900	1900

	m_n^*/m_0	m_p^*/m_0
Silicon	1.08	0.56
Gallium Arsenide	0.067	0.48
Germanium	0.55	0.37

$$\mu_n = \frac{e\tau_{cn}}{m_n^*}$$

Scattering

1. **Phonon scattering** - Lattice atoms vibrate in discrete allowable states (quantum mechanics). This can be modeled as discrete particles called phonons. Transfer of energy from electrons to the lattice is called phonon scattering.
2. **Ionized impurity** atom scattering (important at high dopant concentrations).
3. **Surface/interface scattering** (important in MOS devices).
4. **Neutral impurity** atom scattering (usually negligible). E.g Low T.
5. **Electron - electron** and electron - hole scattering (important at high carrier concentrations).
6. **Crystal defects** (important in polycrystalline material).

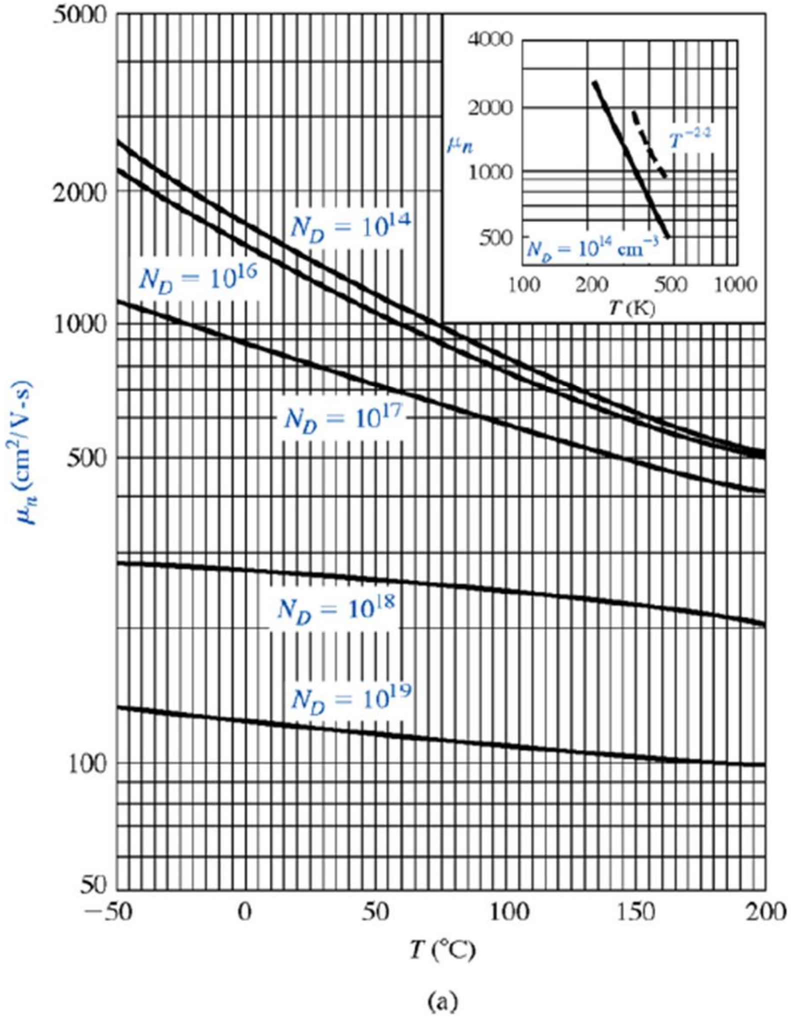
Phonon Scattering

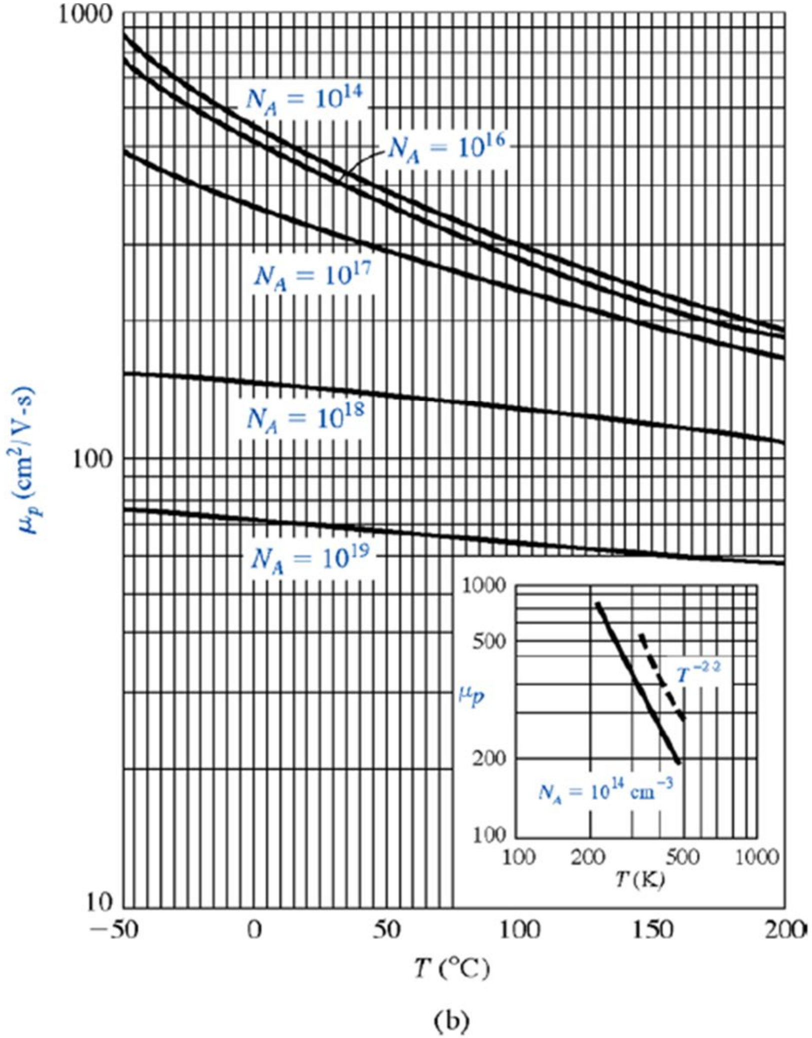
Phonons are lattice vibrations (Atoms randomly vibrate about their position @ T>0K)

Lattice vibration causes a local volume change and hence lattice constant change. → Bandgap generally widens with a smaller lattice constant. → Disruption of valence and conduction band edges scatters the carriers.

•Mobility due to lattice scattering $\mu_L \propto T^{-\frac{3}{2}}$ (as temp increases vibration of atoms also increases)

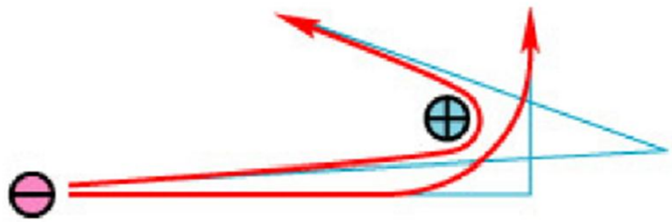
Phonon Scattering



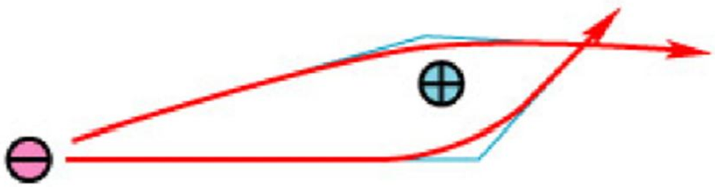


(a) Electron and (b) Hole mobilities in Si vs. T at different doping concentrations. (Inserts show dependence for almost intrinsic Si)

Ionized impurity scattering



Low electron velocities



High electron velocities

Scattering due to coulomb interaction between electrons/holes and ionized impurities.

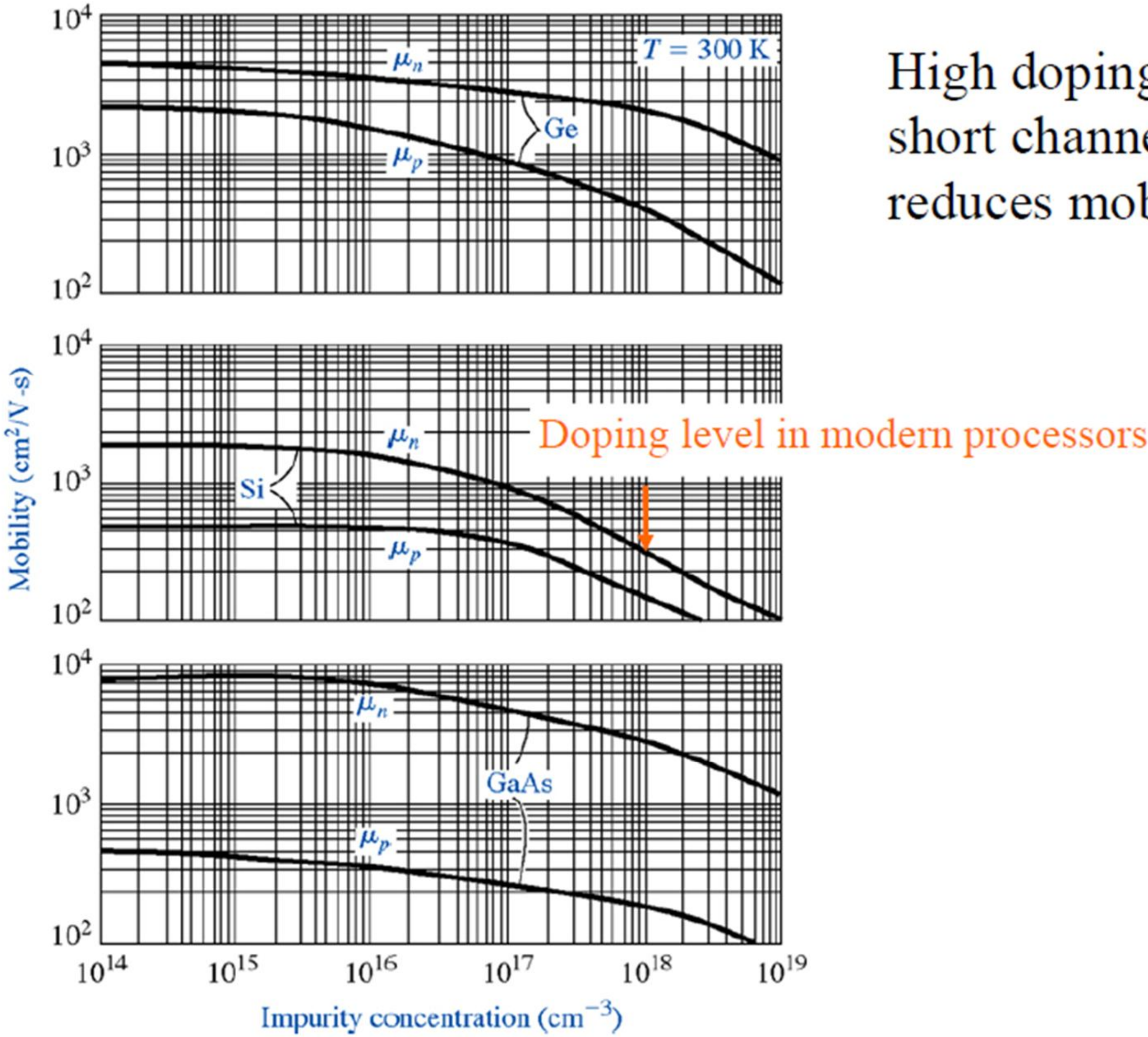
$$\mu_I \propto \frac{T^{+3/2}}{N_I}$$

T increases → thermal velocity v_{th} increases, so less time spent for scattering → $\mu_I \propto T^n$

$$N_I = N_d^+ + N_a^-$$

N_I increases → the scattering chance increases → $\mu_I \propto 1/N_I$

Ionized impurity scattering



High doping is required to overcome short channel effects even though it reduces mobility.

Electron and Hole mobility vs. impurity concentration.

How to combine mobility effects?

τ_I : average time between two collisions with dopant atoms

τ_L : average time between two collisions with "vibrating" lattice points

$\frac{dt}{\tau_I}$: Number of collisions in time dt due to impurity scattering

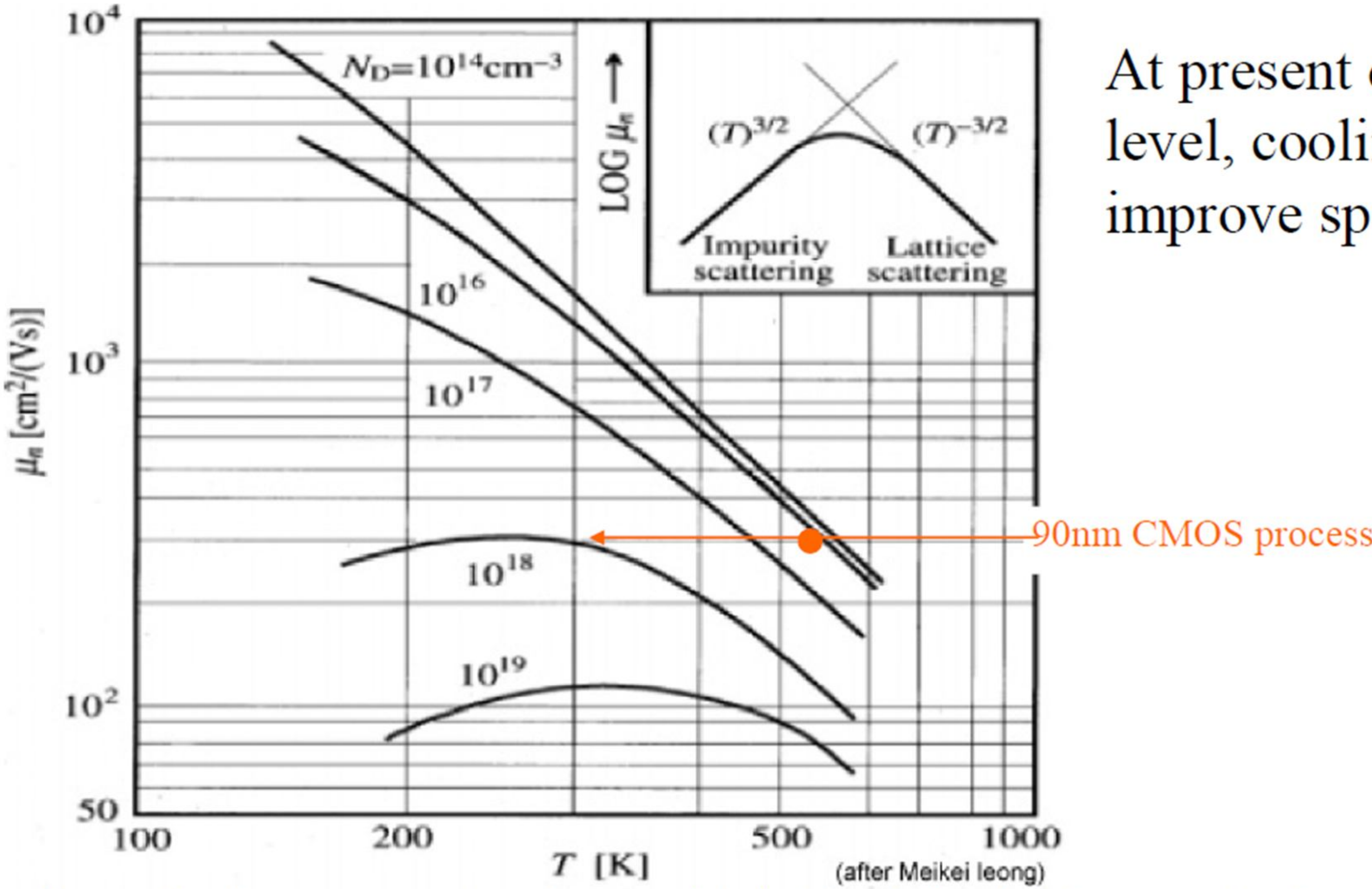
$\frac{dt}{\tau_L}$: Number of collisions in time dt due to lattice scattering

Total number of collisions in dt: $\frac{dt}{\tau} = \frac{dt}{\tau_I} + \frac{dt}{\tau_L}$

$$\mu_n = \frac{e\tau_{cn}}{m_n^*}$$

$$\boxed{\frac{1}{\mu} = \frac{1}{\mu_I} + \frac{1}{\mu_L}}$$

Electron mobility of Si vs. T for various Nd



At present doping level, cooling does not improve speed much.

Low doping concentration or high T → The lattice scattering dominates

High doping concentration or low T → The impurity scattering dominates

Conductivity

$$J_{drf} = e(\mu_n n + \mu_p p)\mathrm{E} = \sigma \mathrm{E}$$

$$\boxed{\sigma = e\mu_n n + e\mu_p p}$$

$\sigma \rightarrow$ *conductivity*

Units $\rightarrow (\Omega\text{-cm})^{-1}$

$$\boxed{\rho = \frac{1}{\sigma} = \frac{1}{e(\mu_n n + \mu_p p)}}$$

$\rho \rightarrow$ *resistivity*

Units $\rightarrow \Omega\text{-cm}$

Conductivity

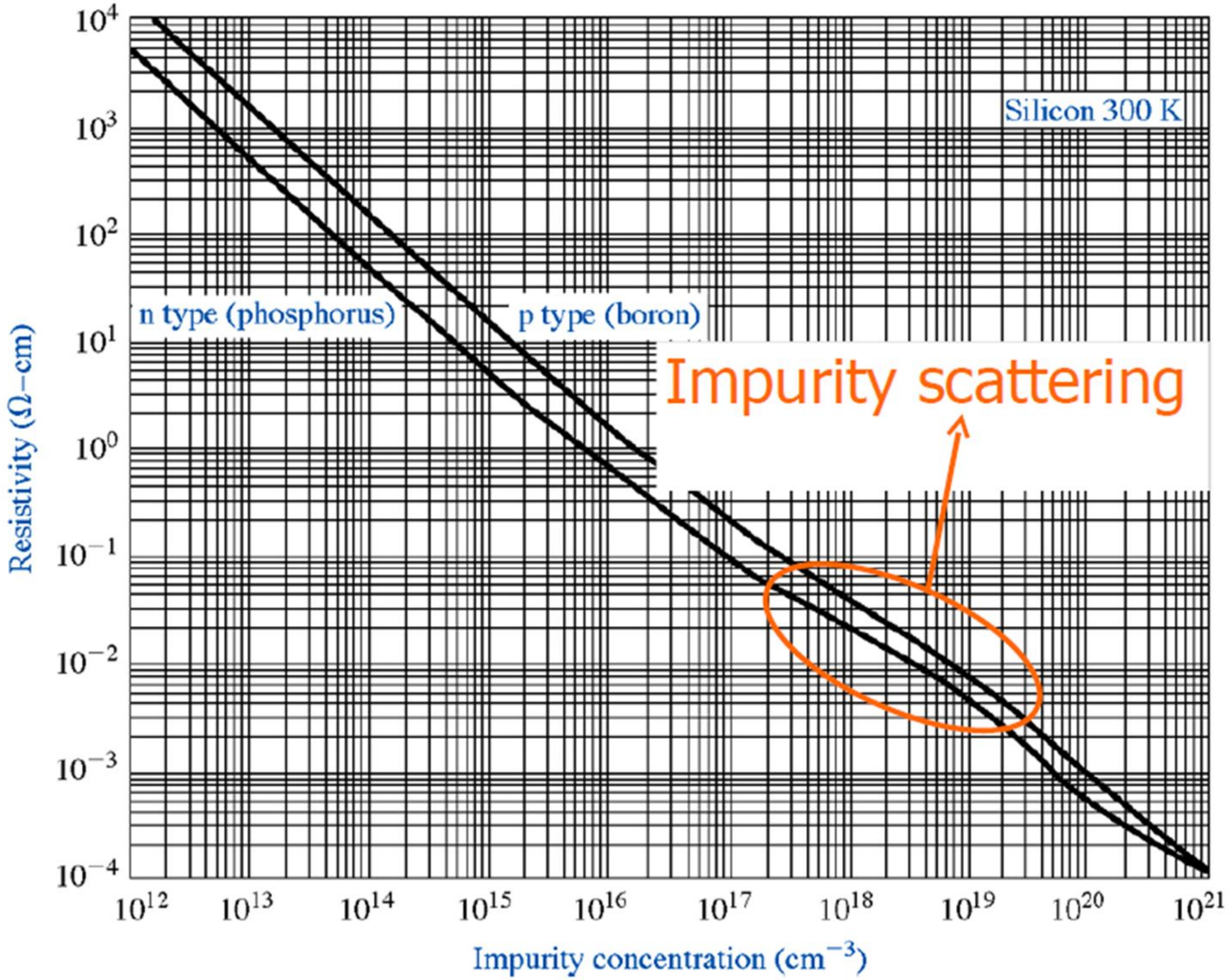
If we assume complete ionization,

$$\sigma = \frac{1}{\rho} = e\mu_n N_d \text{ or } e\mu_p N_a$$

But curve not linear because -

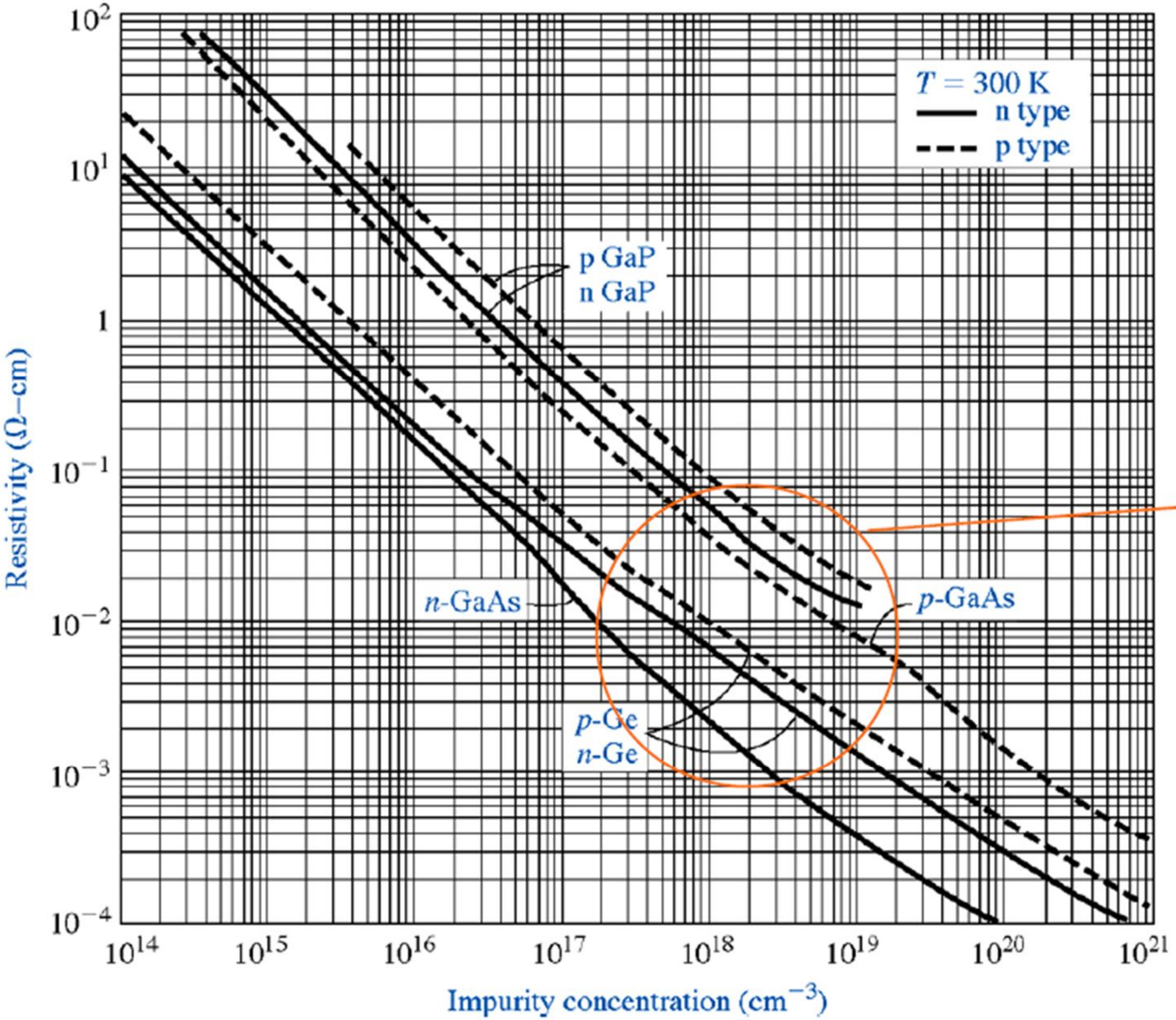
$$\boxed{\frac{1}{\mu} = \frac{1}{\mu_I} + \frac{1}{\mu_L}}$$

Impurity scattering affects mobility.



- Note that resistivity changes by 7 orders of magnitude with doping
- Slope changes from linear to sub-linear ?

Conductivity



Resistivity vs. impurity concentration for Ge, GaAs and GaP

conductivity

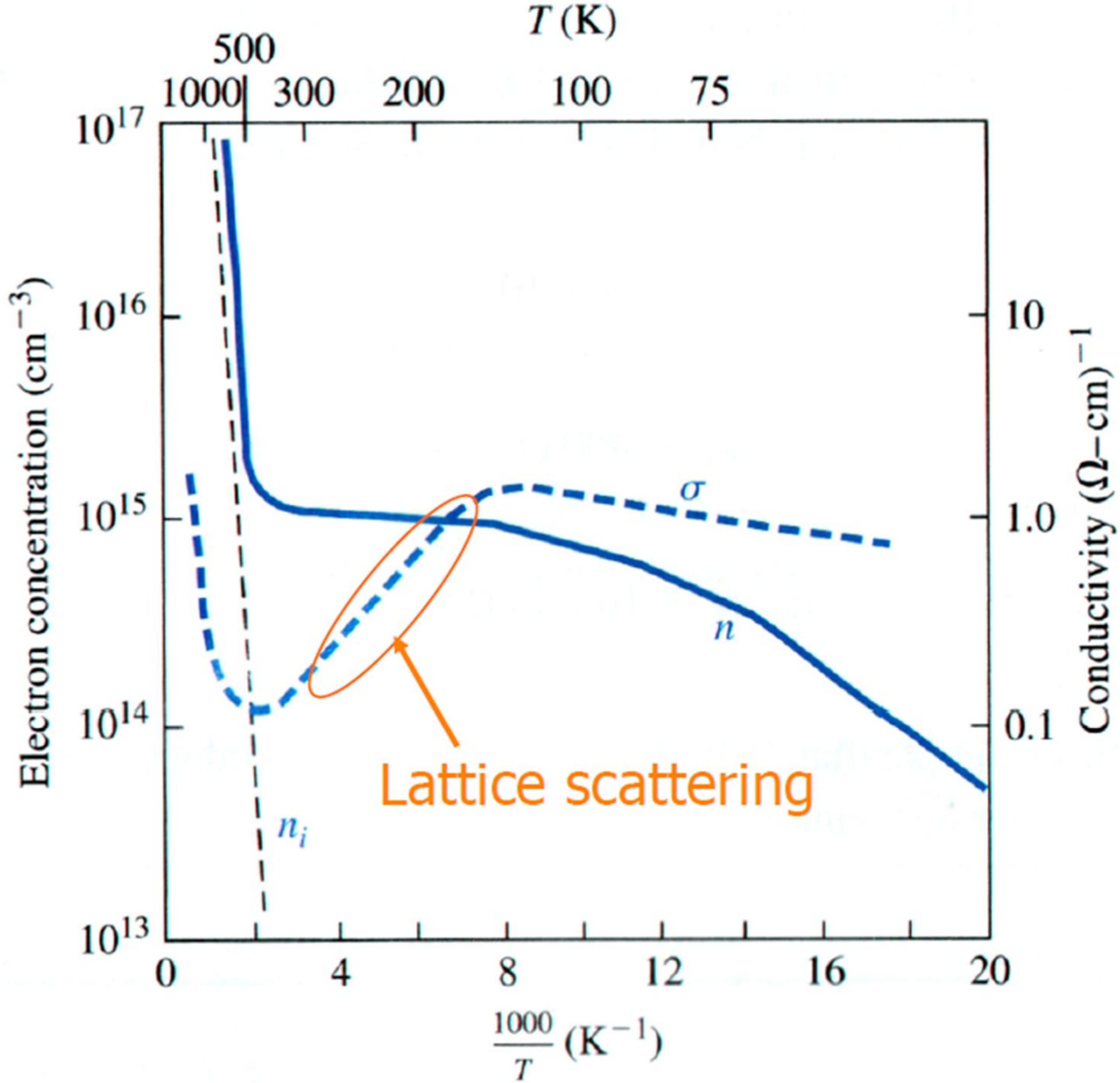
$$\sigma = qn\mu_n + qp\mu_p$$

Impurity scattering affecting mobility.

Questions:
1). Doping always increases the conductivity?

2). Intrinsic semiconductors have the lowest conductivity?

Electron concentration and conductivity vs. 1/T



Electron concentration and conductivity vs. 1/T

- Assuming a n-type material with donor doping N_d >> ni,

$$\sigma = e(\mu_n n + \mu_p p) \approx e\mu_n n \qquad \text{n} \rightarrow \text{electron concentration}$$

- If we also assume complete ionization, $\sigma = 1/\rho = e\mu_n N_d$

Mid temp → complete ionization → n constant at N_d but μ reduces with temp due to lattice scattering → so conductivity drops.

- At high temperature, intrinsic carrier concentration increases and dominates both n and σ

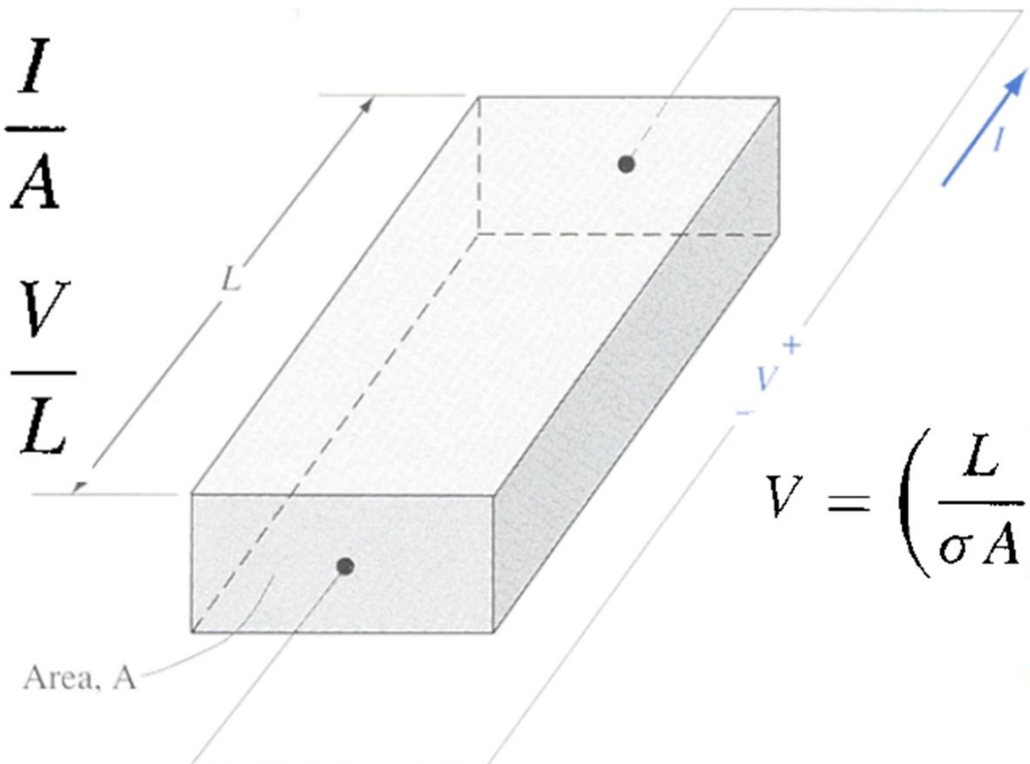
- At low temp, due to freeze-out both n and σ reduce

Ohm's Law

$$J_{drf} = e(\mu_n n + \mu_p p)\mathrm{E} = \sigma \mathrm{E}$$

$$J = \frac{I}{A}$$

$$E = \frac{V}{L}$$



$$\frac{I}{A} = \sigma \left(\frac{V}{L}\right)$$

$$V = \left(\frac{L}{\sigma A}\right) I = \left(\frac{\rho L}{A}\right) I = IR$$

Ohm's law

Velocity saturation

Ohm's "Law"

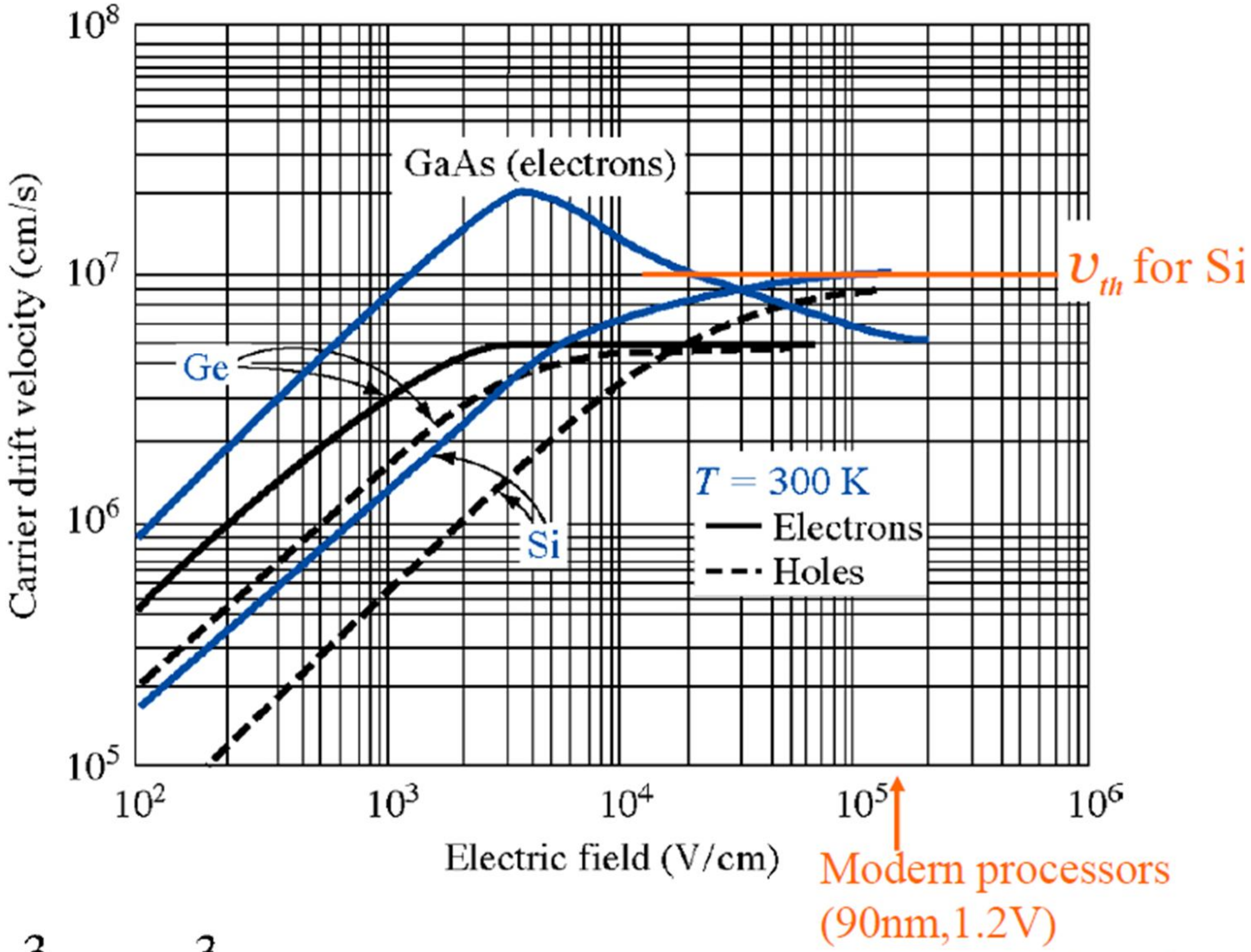
- This says the **Drift Velocity V_d** is **linear in the electric field E:** $$V_d = \mu E$$

$\mu \equiv$ Mobility

- If this were true for all **E**, the charge carriers could be made to go fast without limit, just by increasing **E**! That would be nonsense! So, in every material, ***at high enough*** **E**, the **V_d vs E curve** ***must saturate to a constant value!***

Velocity saturation

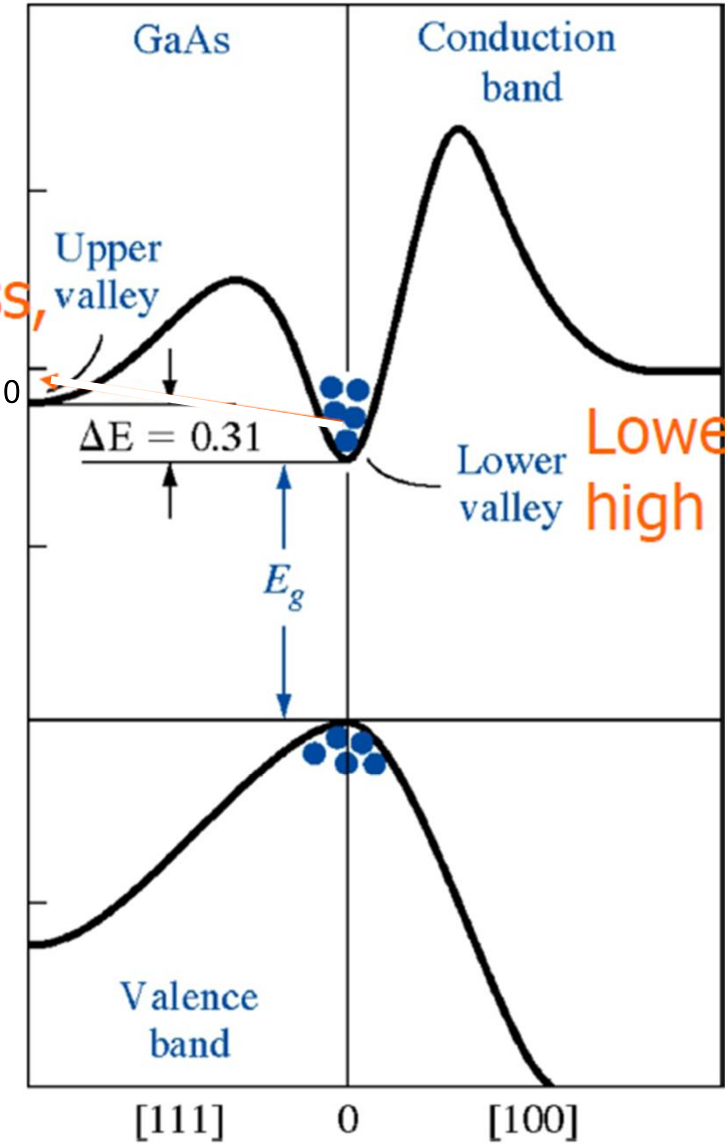
$$\boxed{\upsilon_d = \mu E}$$



Random thermal energy $= \frac{1}{2} m \upsilon_{th}^2 = \frac{3}{2} kT = \frac{3}{2}(0.0259) = 0.03885 \text{ eV} \longrightarrow V_{th} \approx 10^7 \text{cm/s}$

Intervalley transfer mechanism in GaAs

Higher effective mass, low mobility $m^*=0.55m_0$



Lower effective mass, high mobility $m^*=0.067m_0$

At higher E → lower mobility → lower current → negative resistance

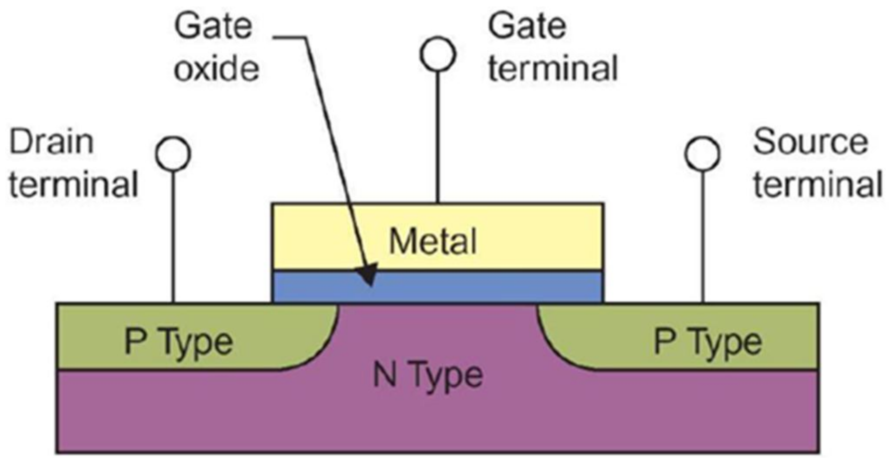
Why velocity is saturated at large E?

For a certain semiconductor, mean free path L is a constant, for example, for Si, L~=1 um, and thus mean free time :

τ_{cp} =1 um/V_{th}=1 psec, where V_{th}=10^7 cm/sec is the thermal motion velocity.

As E increased, V= uE> V_{th}, then τ= L /V< τ_{cp} and thus *u< u_0, which in turn reduces the increasing of V (because V=uE, u is reduced when E is increased to a certain limit).*

Mobility: valid at short lengths ?



Mean Free Path (l_m)
~ 10 nm to 1μm

E_c = 2 V/μm

32nm transistor

L_G = 30nm

V_{DD} = 1V

E = 1V/30nm = 33 V/μm

Ballastic transport becomes important in nanoscale transistors

Velocity saturation

It is important to note that the concept of velocity-field relation is valid if the fields are changing slowly over distances comparable the electron mean free path. This is the case in devices that are longer than a micron or so. For sub-micron devices electrons can move without scattering for a some distance. In this case the transport is called ballistic transport and is described by the Newton's equation without scattering,

For short distances electrons can display ***overshoot effects*** i.e they can have velocities larger than what may be expected from a steady state velocity-field relation. For light mass semiconductors such as GaAs and InGaAs velocity overshoot effects dominate modern devices.

Channel length < mean free path:
scattering can be neglected and ballistic transport dominates

$$J = nqv = nquE$$

$$\mu_n = \frac{e\tau_{cn}}{m_n^*}$$

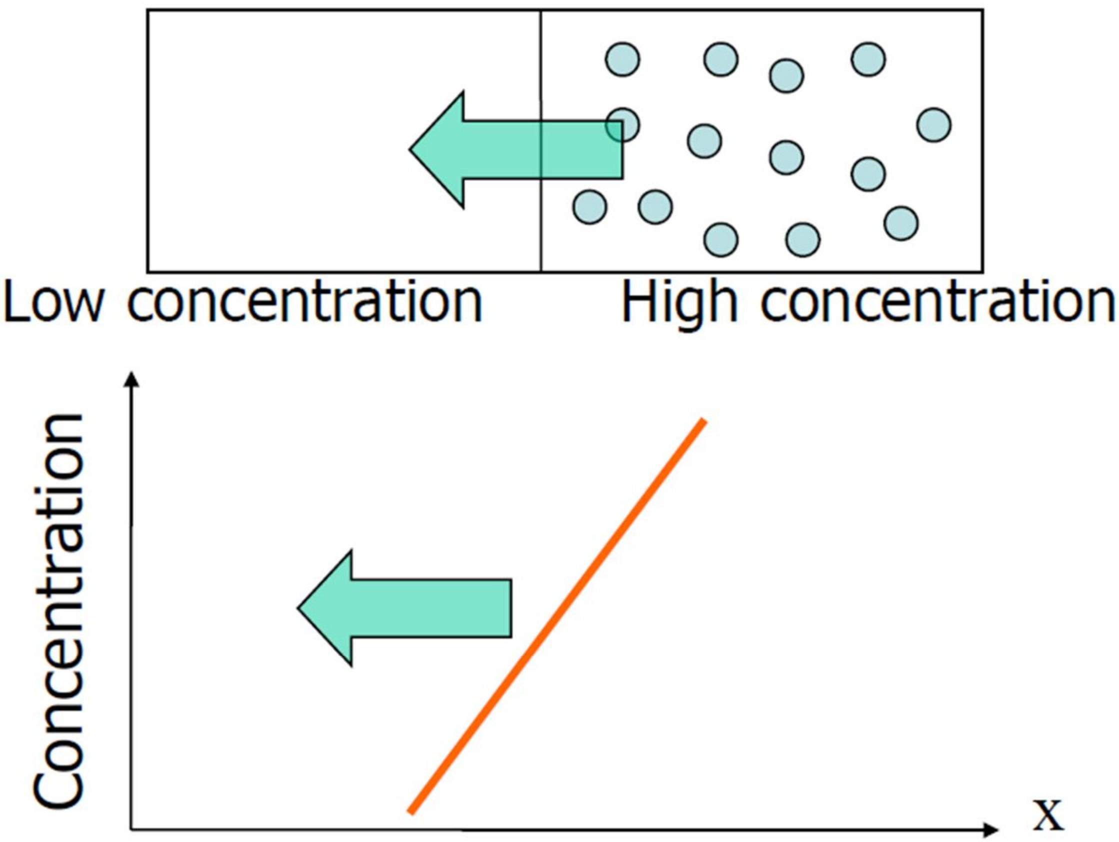
$$\boxed{\frac{1}{\mu} = \frac{1}{\mu_I} + \frac{1}{\mu_L}}$$

$$\sigma = e\mu_n n + e\mu_p p$$

Velocity saturation $V_d = \mu E$

Ballistic transport

5.2 | CARRIER DIFFUSION



Positive slope in x-direction → a flux towards negative-x direction

Diffusion current density

for holes

$$\boxed{J_{px|dif} = -eD_p \frac{dp}{dx}}$$

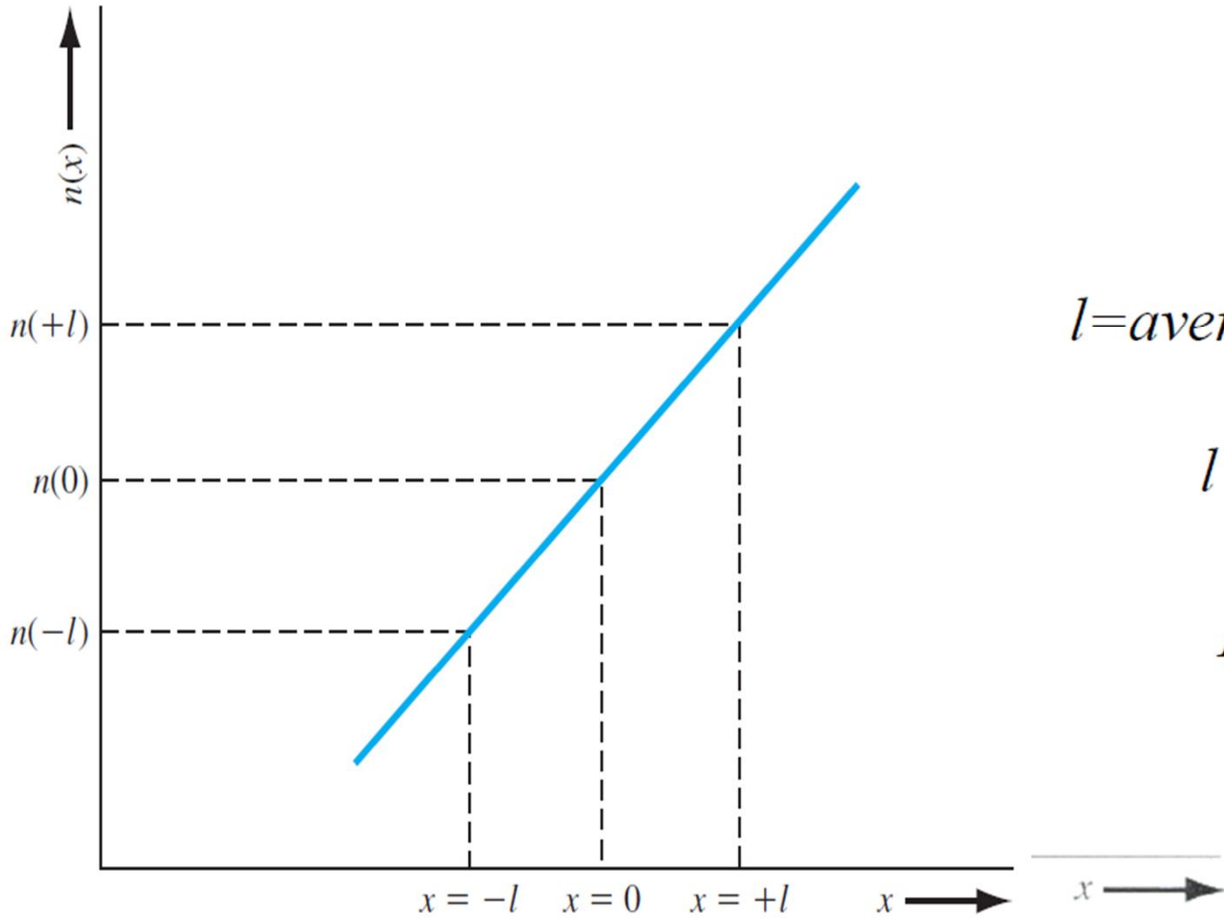
D: diffusion coefficient

for electrons

$$\boxed{J_{nx|dif} = eD_n \frac{dn}{dx}}$$

$$J_{(n,p)x|dif} = eD_n \frac{dn}{dx} - eD_p \frac{dp}{dx}$$

Diffusion coefficient (I)



Electron concentration versus distance.

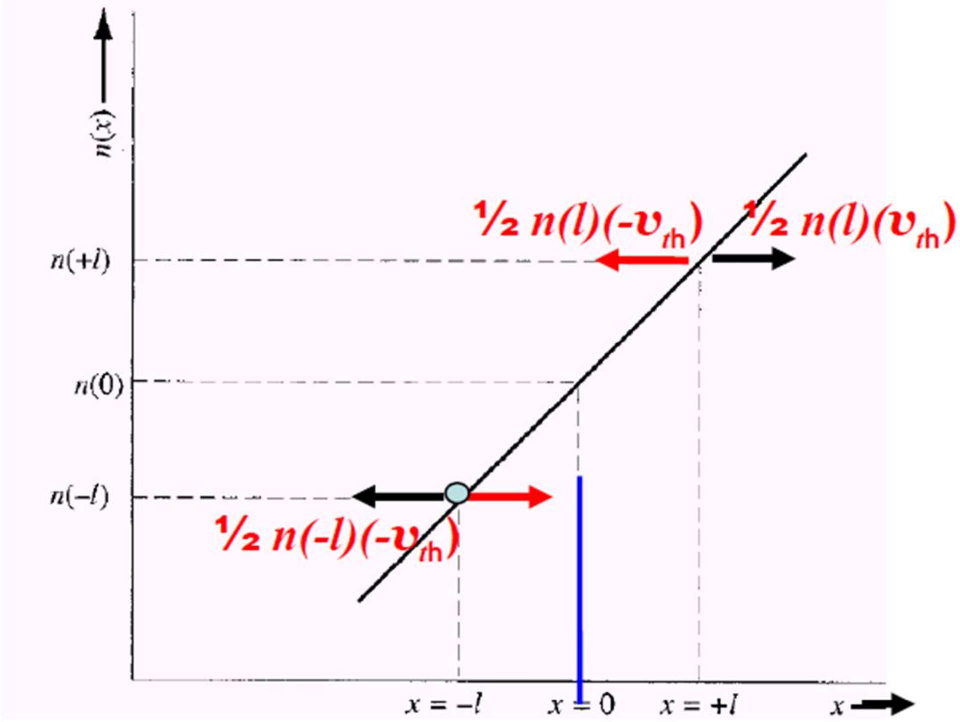
l=average mean free path

$$l = (v_{th} + v_{dr})\,\tau_{cp}$$

If $E=0$, $vdr=0$

$$l = v_{th}\,\tau_{cp}$$

Diffusion coefficient (II)



F_n →net rate of electron flow in the $+x$ direction at $x = 0$

= sum of electron flow in +x direction at x=-l and electron flow in –x direction at x=l

$$F_n = \frac{1}{2}n(-l)\upsilon_{th} - \frac{1}{2}n(+l)\upsilon_{th} = \frac{1}{2}\upsilon_{th}[n(-l) - n(+l)]$$

Diffusion coefficient (III)

$$F_n = \frac{1}{2}n(-l)\upsilon_{th} - \frac{1}{2}n(+l)\upsilon_{th} = \frac{1}{2}\upsilon_{th}[n(-l) - n(+l)]$$

Taylor expansion of $n(-l)$ en $n(+l)$ at $x = 0$:

$$n(+l) = n(0) + l\,\frac{dn}{dx} + \ldots\ldots$$

$$F_n = \frac{1}{2}\upsilon_{th}\left\{\left[n(0) - l\,\frac{dn}{dx}\right] - \left[n(0) + l\,\frac{dn}{dx}\right]\right\}$$

$$F_n = -\upsilon_{th} l\,\frac{dn}{dx}$$

Diffusion coefficient (IV)

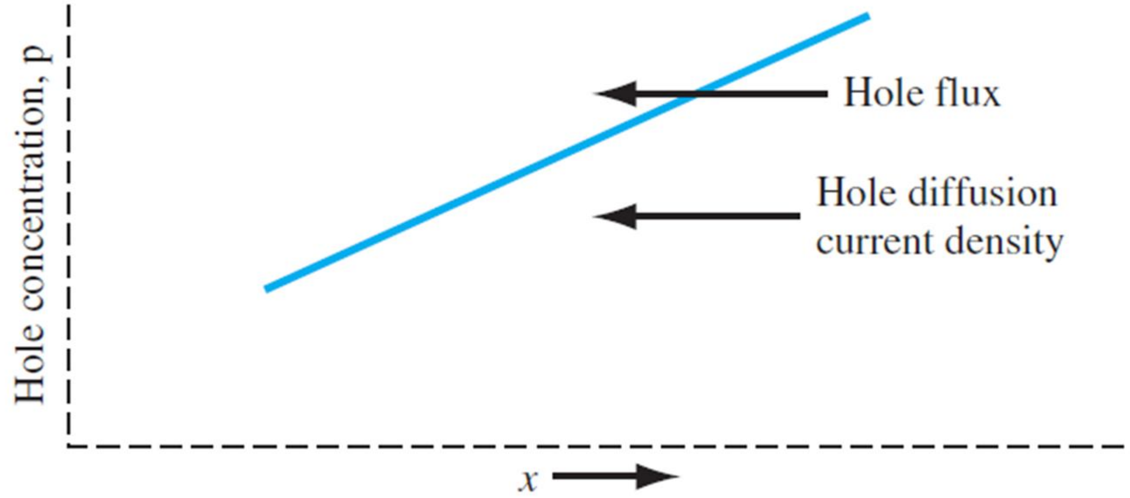
Recall that $F_n = -\upsilon_{th} l \dfrac{dn}{dx}$

The current is $J = -eF_n = +e\upsilon_{th} l \dfrac{dn}{dx}$

D: diffusion coefficient = $v_{th} l$ (cm^2/s)=$v_{th}^2\ \tau_{cp}$

$$\boxed{J_{nx|dif} = eD_n \frac{dn}{dx}}$$

For holes



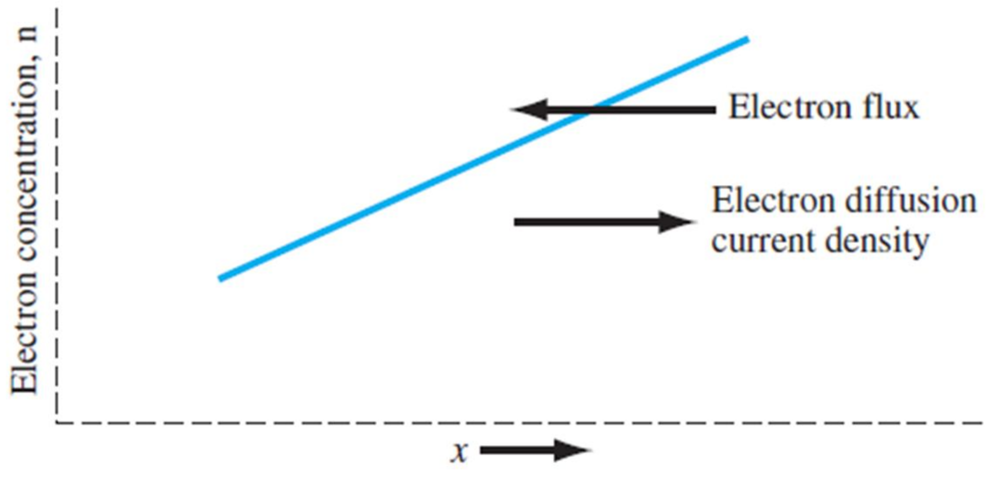
$$\boxed{J_{px|dif} = -eD_p \frac{dp}{dx}}$$

Note

D_p → *hole diffusion coefficient*

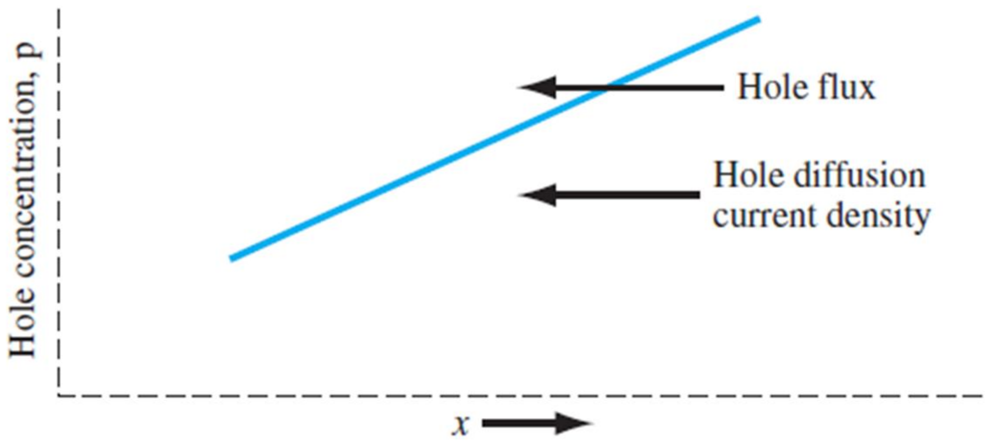
$J_{px|dif}$ → hole diffusion current density for one-dimensional case

Diffusion current



(a)

$$\boxed{J_{nx|dif} = eD_n \frac{dn}{dx}}$$



(b)

$$\boxed{J_{px|dif} = -eD_p \frac{dp}{dx}}$$

Total Current Density

$$J = \boxed{en\mu_n \mathrm{E}_x + ep\mu_p \mathrm{E}_x} + \boxed{eD_n \frac{dn}{dx} - eD_p \frac{dp}{dx}}$$

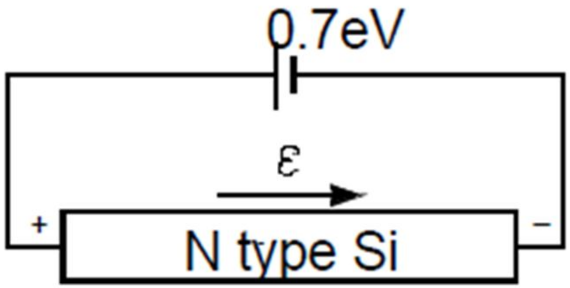
DRIFT

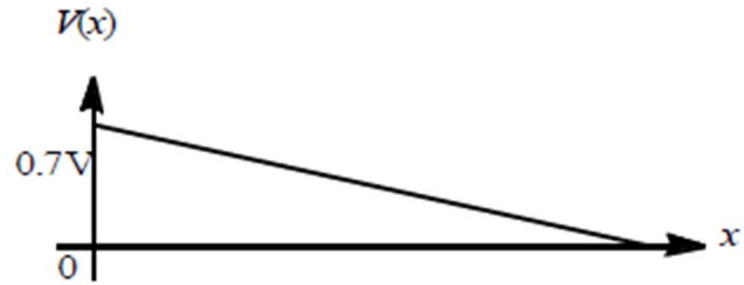
DIFFUSION

Generalized current Density Equation -

$$J = en\mu_n \mathrm{E} + ep\mu_p \mathrm{E} + eD_n \nabla n - eD_p \nabla p$$

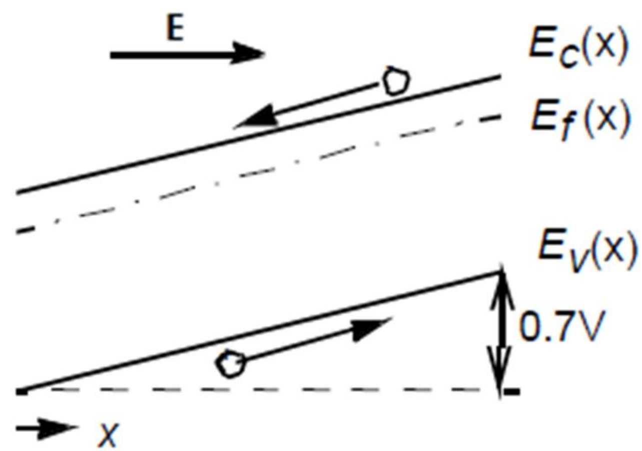
Relation Between the Energy Diagram and V, E





E_c and E_v vary in the opposite direction from the voltage. That is, E_c and E_v are higher where the voltage is lower.

$$\mathbf{E}(x) = -\frac{dV}{dx} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_v}{dx}$$

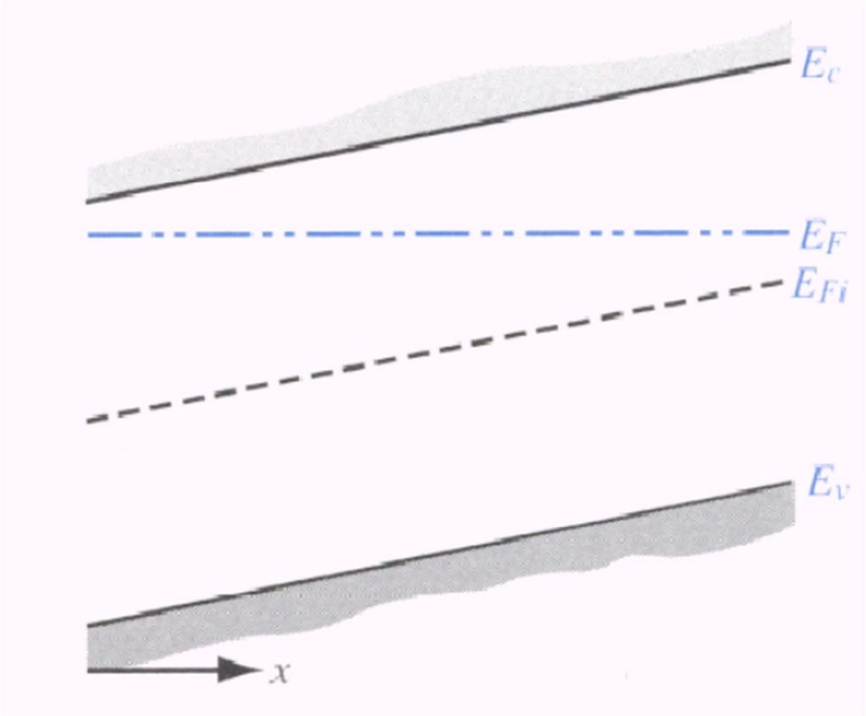


Electrical field E= - dV/dx
Electrical energy E_c= - qV

能带的高度**=>**电压
能带的斜率**=>**电场
电场的斜率**=>**电荷

5.3 | GRADED IMPURITY DISTRIBUTION

Consider a case where donor concentration increases in -x direction



Diffusion of electrons

↓

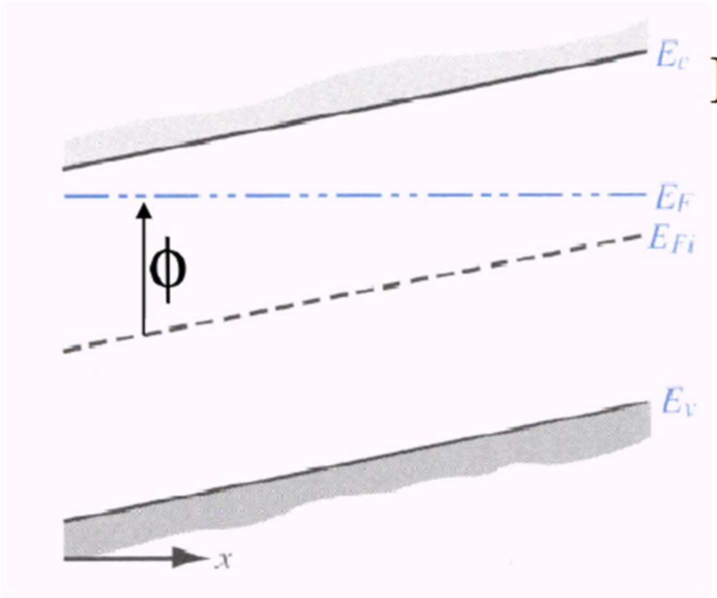
Formation of electric field

↓

Prevents further diffusion

- Due to electric field → potential difference across the device
- In the region at lower potential → E_F-E_{Fi} higher

Induced Electric Field Ex



Potential: $\phi = +\frac{1}{e}(E_F - E_{Fi})$

Electric field

$$E_x = -\frac{d\phi}{dx} = \frac{1}{e}\frac{dE_{Fi}}{dx}$$

Assuming Electron concentration $\approx$ Donor concentration

$$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right] \approx N_d(x)$$

Induced Electric Field Ex

Recall $$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right] \approx N_d(x)$$

Taking log $$E_F - E_{Fi} = kT \ln\left(\frac{N_d(x)}{n_i}\right)$$

Take the derivative with respect to x $$-\frac{dE_{Fi}}{dx} = \frac{kT}{N_d(x)} \frac{dN_d(x)}{dx}$$

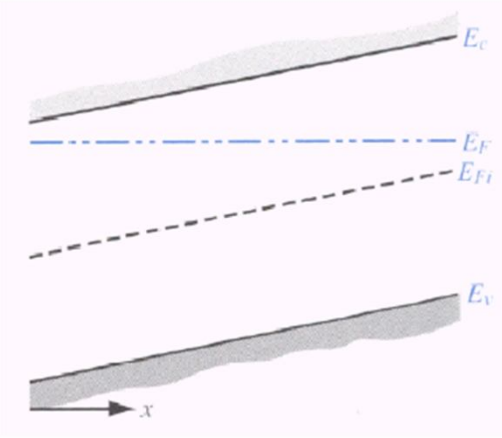
Electric field $$E_x = -\left(\frac{kT}{e}\right) \frac{1}{N_d(x)} \frac{dN_d(x)}{dx}$$

The Einstein Relation

Consider the general current density equation:

$$\boxed{J_{n,p} = en\mu_n \mathrm{E}_x + ep\mu_p \mathrm{E}_x + eD_n \frac{dn}{dx} - eD_p \frac{dp}{dx}}$$

Assume n-graded semiconductor material in thermal equilibrium:



$$J_n = 0 = en\mu_n \mathrm{E}_x + eD_n \frac{dn}{dx}$$

Assuming n≈N_d

$$J_n = 0 = e\mu_n N_d(x)\mathrm{E}_x + eD_n \frac{dN_d(x)}{dx}$$

The Einstein Relation

Substituting $E_x = -\left(\frac{kT}{e}\right)\frac{1}{N_d(x)}\frac{dN_d(x)}{dx}$

$$0 = -e\mu_n N_d(x)\left(\frac{kT}{e}\right)\frac{1}{N_d(x)}\frac{dN_d(x)}{dx} + eD_n\frac{dN_d(x)}{dx}$$

$$0 = \left[-e\mu_n\left(\frac{kT}{e}\right) + eD_n\right]\frac{dN_d(x)}{dx}$$

$$\frac{D_n}{\mu_n} = \frac{kT}{e}$$

because $\frac{dN_d}{dx} \neq 0$

The Einstein Relation

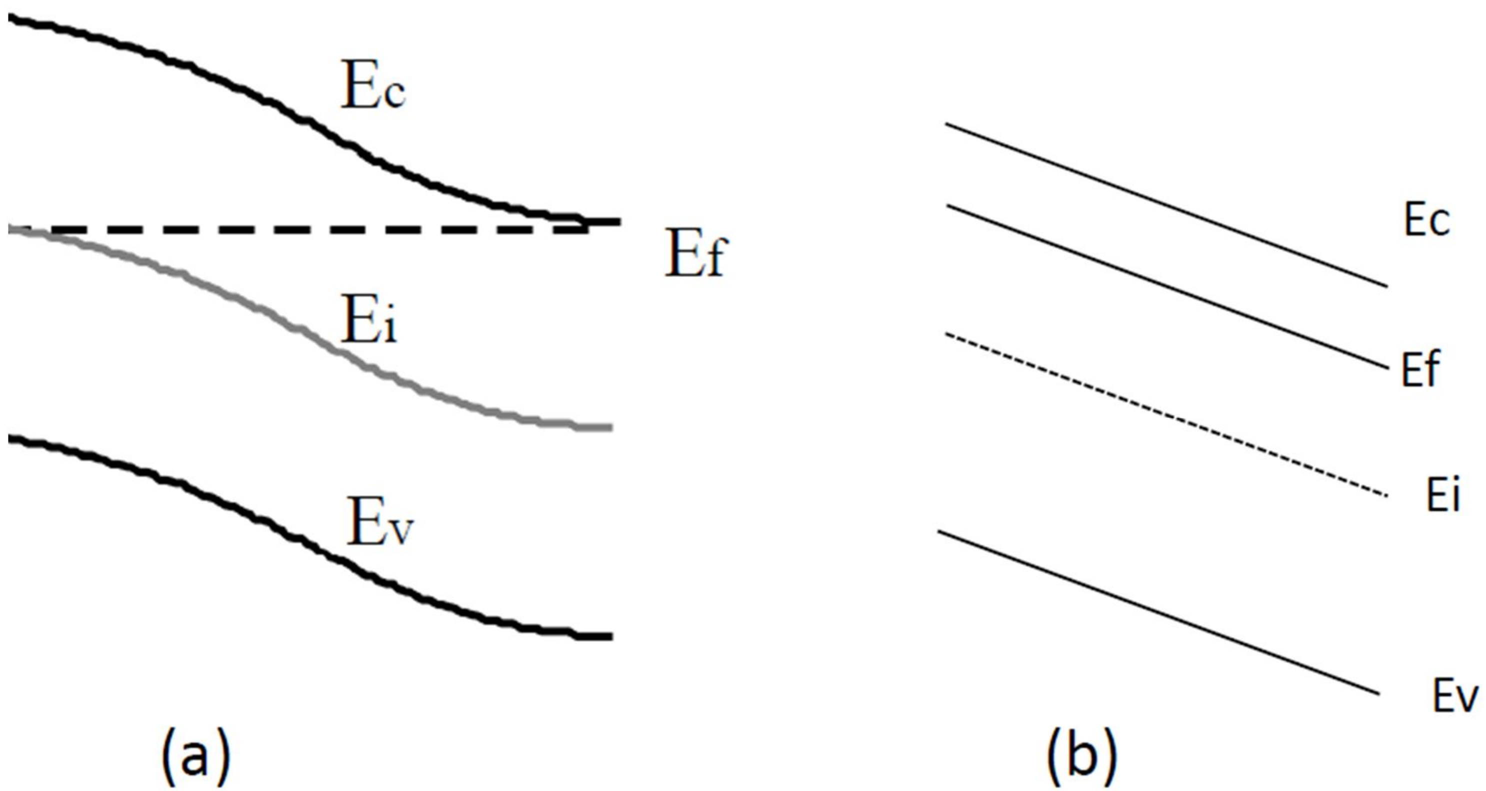
$$\frac{D_n}{\mu_n} = \frac{kT}{e}$$

Similarly $$\frac{D_p}{\mu_p} = \frac{kT}{e}$$

Einstein relation $\longrightarrow$ $$\boxed{\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}}$$

Question

For below band diagrams, what are the differences between them?
Specify the differences in terms of doping, electric field, equilibrium etc.



*5.4 | THE HALL EFFECT

- n- or p-type, carrier concentration and mobility can be experimentally measured.
- Electric and magnetic fields are applied to a semiconductor.

The Hall Effect

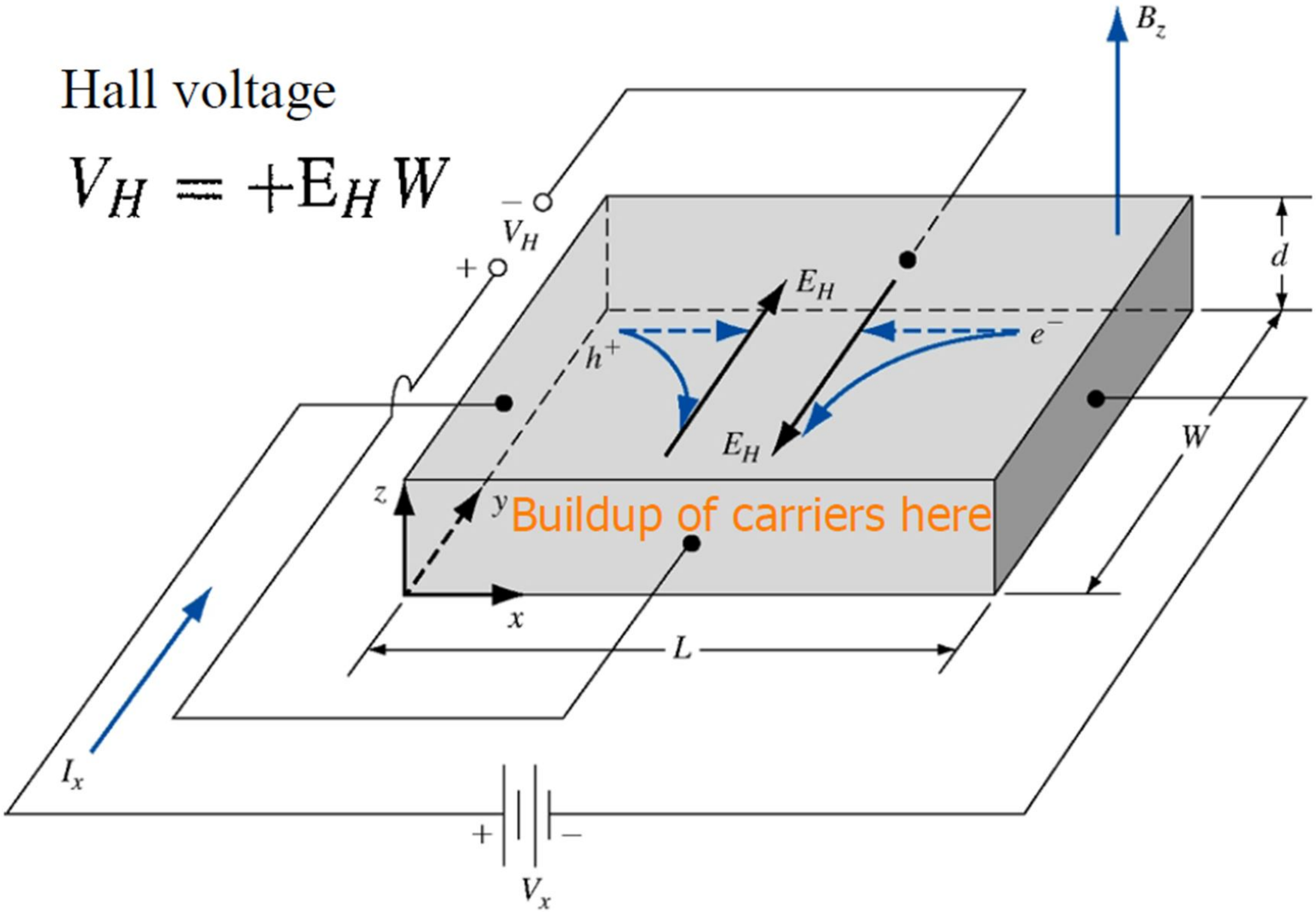
The Hall Effect

(Lorentz force)

$F = qv_x \times B_z$

Hall voltage

$$V_H = +E_H W$$



The Hall Effect

- Due to magnetic field, both electrons and holes experience a force in –y direction.

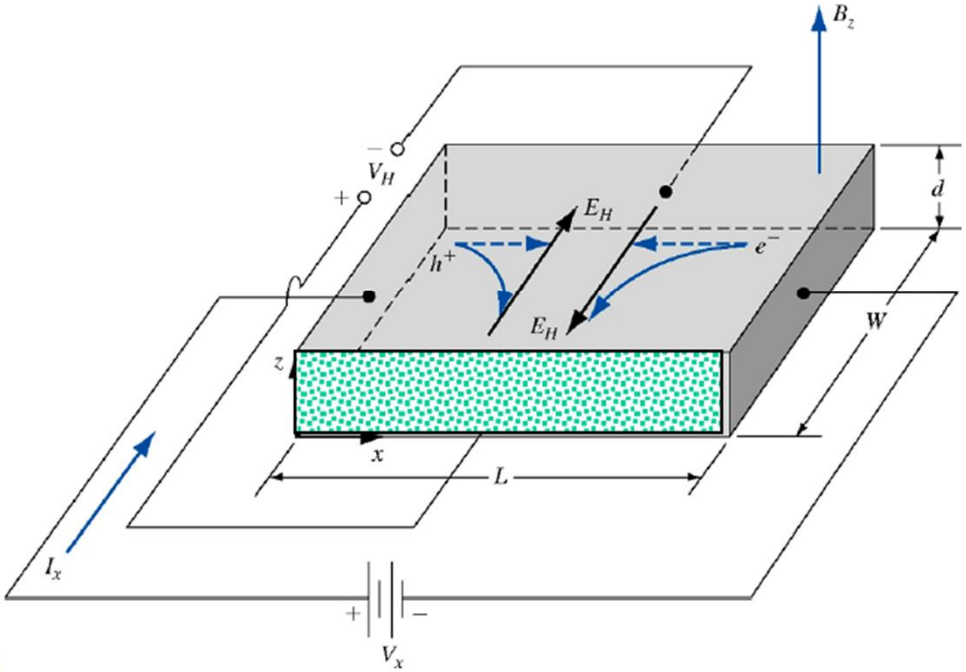
- In n-type material, there will be a build up of –ve charge at y=0 and in p-type material, there will be a build up of +ve charge.

- The net charge induces a electric force in y-direction opposing the magnetic field force and in steady state they will exactly cancel each other.

Hall Voltage

Hall Voltage:

$$F = q[\mathrm{E} + \upsilon \times B] = 0$$



$$q\mathrm{E}_y = q\upsilon_x B_z \longrightarrow \mathrm{E}_y = \mathrm{v}_x\mathrm{B}_z$$

Hall voltage

$$\mathrm{V}_\mathrm{H} = \mathrm{E}_y\mathrm{W} = \mathrm{v}_x\mathrm{W}\mathrm{B}_z$$

How to determine n-type or p-type & doping concentration?

$$V_H = E_y W = v_x W B_z$$

For p-type material: V_H will be +ve

$$v_x = \frac{J_x}{ep} = \frac{I_x}{(ep)(Wd)} \rightarrow V_H = \frac{I_x B_z}{epd} \rightarrow \boxed{p = \frac{I_x B_z}{edV_H}}$$

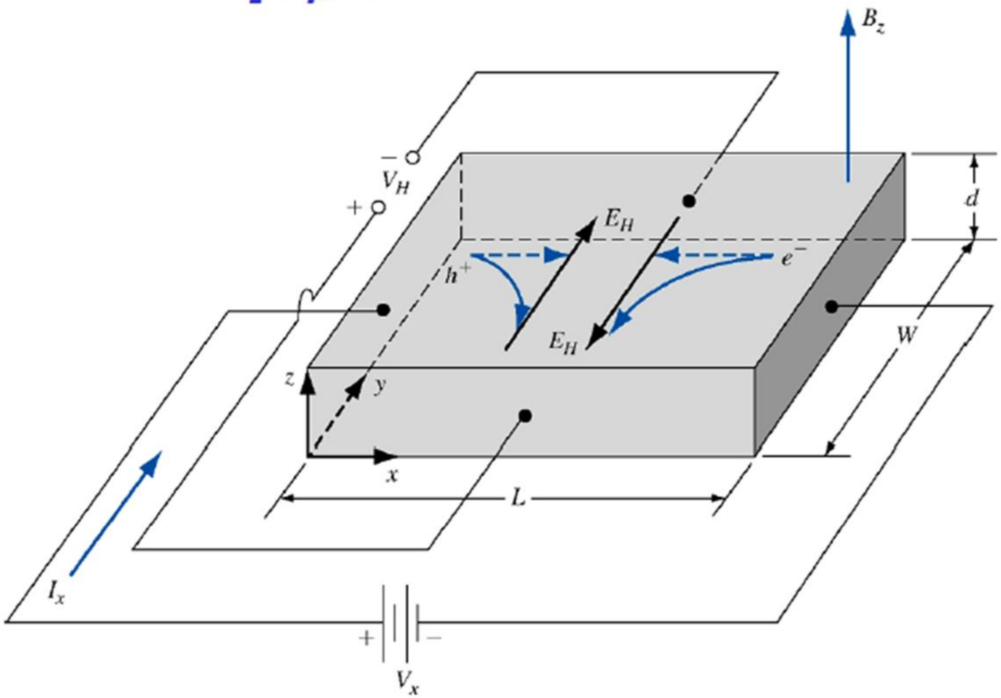
For n-type material: V_H will be -ve

$$V_H = -\frac{I_x B_z}{ned} \longrightarrow \boxed{n = -\frac{I_x B_z}{edV_H}}$$

For n-type material V_H will be -ve → n will still be +ve !

How can you determine μ_n

How can you determine μ_n, μ_p?



$$J_x = ep\mu_p \mathrm{E}_x$$

$$\frac{I_x}{Wd} = \frac{ep\mu_p V_x}{L}$$

$$\boxed{\mu_p = \frac{I_x L}{epV_x Wd} \qquad \mu_n = \frac{I_x L}{enV_x Wd}}$$

Summary

- Drift → net movement of charge due to electric field.
    - Drift current = $J_{drf} = e(n\mu_n + p\mu_p)E$
- Mobility (μ) ← lattice and impurity scattering
    - Mobility due to lattice scattering $\mu_L \propto T^{-\frac{3}{2}}$
    - Mobility due to impurity scattering $\mu_I \propto \frac{T^{\frac{3}{2}}}{N_I}$
- Ohms law → $V = \frac{L}{\sigma A} I = \frac{\rho L}{A} I = RI$
- Drift velocity saturates at high electric field → velocity saturation

Summary

- Some semiconductors → mobility μ reduces at high E → negative resistance → used in design of oscillators.

- Diffusion current $J_{diff} = eD_n \frac{dn}{dx} \quad eD_p \frac{dp}{dx}$

- Einstein relation → relation between diffusion coefficient and mobility → $\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$

- Hall effect → can be used to determine semiconductor type, doping concentration and mobility

Reading Materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 5, “Carrier Transport Phenomena”.